

The Social Radars

S3: Amjad Masad & Haya Odeh, Co-Founders of Replit

Released August 14 , 2024

Jessica: I'm Jessica Livingston. And Carolynn Levy and I are The Social Radars. In this podcast, we talk to some of the most successful founders in Silicon Valley about how they did it. Carolynn and I have been working together to help thousands of startups at Y Combinator for almost 20 years. Come be a fly on the wall as we talk to founders and learn their true stories.

Carolynn, I'm really excited today we have Haya Odeh and Amjad Masad, who are the founders of Replit.

Carolynn: Very exciting.

Jessica: And Replit went through YC in 2018 and it's basically programming and hosting at the click of a button environment. But first of all, welcome you two, and I would like you two to describe to our listeners what Replit does.

Amjad: Thank you so much for having us. This is great. So yeah, what Replit is, I mean, your description is great. Actually, I might start using it as this one-liner. But basically, historically programming has just been very difficult to start with. You have to install loads of software, gigabytes of software to even write a single line of code, and that made it really hard to learn, which is why people go to a five-year CS and engineering in school to actually become programmers. It wasn't always that way. And-

Jessica: Really?

Amjad: Yeah, PG talks a lot about Lisp and those days were actually easier because you could just start one program, Lisp, and you can just start coding immediately. But things have gotten more complicated over the years through the 90s and 2000s. And so programming has actually gotten harder to get started with.

And so Replit is in a lot of ways a return to the past where you can just go to a website and the website presents you with a console where you can just type your code and in seconds you're programming.

And then we took the same mindset to every aspect of the software development lifecycle, which is, what if you want to collaborate with someone, and typically do it via Git and that's also very complicated? With Replit it's like Google Docs, I'll send you a link and you'll hop into my environment and I'll see your cursor. It's really fun.

And then what if you want to host something? Well, that's also fairly complicated. You want to go to AWS and they're going to present you with a lot of crap, to be honest. And

so we've made that part easier, and so we're taking the same approach across the stack, and that's been very successful.

Jessica: Well, for us non-technical people, Carol and me, I have you filed in my brain as easy programming, so that's why I'm glad that you-

Amjad: Absolutely-

Jessica: Just described it. But I did look at your original application and I did see your mission was to make programming more accessible and you dreamed of, rather than creating unique programming environments, every time you wanted to program, you just wanted to open a tab. And so that's what Replit's done, and I love that.

Will you take me back to how it got started? Because I was surprised to see you had applied to Y Combinator before doing an educational... like teaching for teachers. Tell us the story of how Replit came to be.

Amjad: Yeah, so Replit has a very long history and backstory. It started in college. I was going to school back in Jordan where Haya and I are from, and I didn't have a laptop at the time, and I was running into all the problems we just talked about, which is every time I want to learn a new programming language, I have to do the same song and dance. And it was pretty difficult. So I was like, "Why isn't there a website you can go to and write code?" And there wasn't and that was very surprising. So that was like 2008, 2009. And it was profoundly surprising to me because you would expect a good idea like that to have people working on it. And I thought, "How hard could it be?"

So I started working on it and had a bit of an immediate success in terms of, I put a few toy programming languages on the browser. I was reading PGs at the time, essays, and I was reading about Lisp and getting excited about that, reading Hacker News. And so I put Lisp there because Lisp, it's easy to make Lisp and my friends liked it. They really, really liked it on the first try. And I thought, "Okay, this idea has legs. And so okay, let me go add a real programming language, like Python."

Jessica: I think Paul might be a little offended, "A real programming language." Were you-

Amjad: That's the reason-

Jessica: Were you a Lisp hacker or no-

Amjad: I must just add... No, I would say I was into Scheme a little more than Lisp. I just found it to be a bit more pure. I was into it as a toy language. I was never into it as I'm going to build things with Lisp. It's a great sandbox to play with ideas around languages, but to me it stopped being practical for a long time. Don't tell this to Paul, so...

Jessica: Don't worry. He'll listen to this podcast.

Amjad: Oh, no. So then the difficulty started and actually took a few years before I had any breakthrough to make the application more useful. To add Python, I had to compile the Python interpreter from C to JavaScript so that it runs in the browser. And we were the first to do that.

And I open sourced all of that and put it on Hacker News in 2011 and it went viral. It was on top of Hacker News for a few days, and everyone on the internet was super excited about it and tweeting about it and all of that.

Jessica: Oh, wow.

Amjad: And that's when I recruited Haya. Haya and I were working together and then dating at the time. And although we don't use dating in Jordan, we're friends. And-

Jessica: Okay.

Amjad: We're old now, so it's fine. And I'll let Haya tell her side of the story, but I just needed the design. Because I was just making this thing that wasn't very attractive or interesting. And so that's how Haya got involved.

Jessica: Because, Haya, you are a graphic designer by training, right? Okay. So tell your side of things now.

Haya: I went to design school, yes. But when I graduated, I didn't know what I'm supposed to do. We barely scratched the surface of what design means. So there's art, there's video, there's marketing, there's sales, there is a lot going on there. So when I graduated, I explored a little bit of branding, a little bit of press. The only thing that resonated with me is web design, and I did not have the language for it. Back then, this didn't make any sense, what they are saying here does not match what they want? So the language and the understanding of, oh, well, they're not matching their value prop, or what is the narrative here? Or what are they selling? What are they trying? So all of this language, we didn't know it. We didn't have it.

And I didn't know that there was something called information architecture or UI. The only thing I knew it was like, that's a bunch of text on a website and I can make it look and feel a lot better. When I shared this problem with Amjad, he told me, "Well, I think you would love a book called Don't Make Me Think." And that was his first recommendation to get into web design. And I started digging more into web design and loving redesigning websites on my own time just to practice. And-

Jessica: Oh, wow-

Haya: And that's like, that's-

Jessica: That's pretty nerdy. Okay, you must have loved it. Okay.

Haya: Yeah. And that's my job. So I finish what I need to do, and I just dabble around with this. And I got really bored in my job. I just finished what I needed to know in one year, and then I quit my job because I loved what Amjad gave me to nerd out on, he gave me everything. It's like, take a logo, take a brand, take the UI and think about this thing. So it was fun. It was a fun project and yeah.

Jessica: So you quit your job, but now how did you guys get over to America? When did that happen in the timeline?

Amjad: YC actually was part of that. I don't know if you know that, Jessica?

Jessica: I thought you worked at Facebook and before that Codecademy.

Amjad: Yes. So Codecademy, my first job, Zach and Ryan were just doing YC and they started using the open source software I wrote for Replit.

Jessica: No way.

Amjad: Yes, yes. And as you might know, Codecademy's idea came from Paul, right? Paul was-

Jessica: I do. From our kitchen, a brainstorming session in our kitchen in Palo Alto, I think, right?

Amjad: Yeah. Yes.

Jessica: Okay.

Amjad: Something like that. And I think it's very interesting because this whole story is intertwined with YC and Paul, right? And so they reach out to me and they're like, Hey, really cool stuff that you're working on. We're working on this thing to actually commercialize a lot of the ideas I had. And I thought education is a great use case and market for this. But I wanted to start my own company and I told them basically F-off, I'm not joining you. I want to start my own thing.

Carolynn: A little aggressive.

Amjad: But we're good friends now. But actually Zach to his credit, Zach Sims flew all the way to Jordan.

Carolynn: Wow.

Amjad: And hung out with me and Haya for a week and tried to-

Jessica: Oh my God-

Amjad: Convince me to join him. I took him to the Dead Sea and we went out and I showed him Jordan and all of that, but the offer wasn't attractive to me. I was like, "I want to be a founder or a co-founder or something close to that." So he kept ramping up the offer. I

was driving him to the airport, I was basically just still, "No." And just before I dropped him off, he upped the offer, doubled it or something like that, and I signed the offer letter in the car.

Carolynn: Wow.

Jessica: No.

Amjad: So he scratched the numbers, put the right numbers. I signed the offer letter in the car and I called Haya. I was like, "Hey, I'm going to New York." And Haya will tell you what she said.

Haya: Well, Amjad, when he put the open source project, he started receiving projects and work offers from all over the place, South Africa-

Jessica: Yeah, I believe that.

Carolynn: Yeah, me too-

Haya: [inaudible 00:12:02], Seattle, everywhere. But then when he got the offer in New York, we sat and thought about it, and then we decided to go to New York. And when Zach came and he did sign the offer, he's like, "Oh, I'm going to New York." I'm like, "You're not going to New York before you propose. No one is leaving."

Jessica: "No one is leaving."

Haya: To New York. I'm not playing here, wasting time-

Jessica: Wait a minute, you're just friends. You're just friends.

Haya: No.

Jessica: So wait-

Haya: I'm taking him-

Jessica: I really need to dig into this. Does he call you on his way back from the airport to say, "I've just signed this deal," or did he wait to tell you in person?

Haya: No, he did not, I think wait in person. I think he called me. He called me, yeah-

Jessica: And so you said, "No one's going to New York without a proposal-"

Haya: No one's going. Yeah. Yes.

Jessica: And what did you say, Amjad?

Amjad: I was shocked because I was 23. I didn't plan to get married anytime soon. No, I wasn't 25. You were 25. Yeah, I was 23. And-

Jessica: Shouldn't be laughing.

Amjad: Yeah, it sort of made sense. I wasn't planning to break up or anything like that. I was like, "Okay. I guess." It was a very rational decision. It wasn't like any of those romantic settings and proposals. And eventually we got romantic, but New York was a very romantic place to be. But at the time I was like, "Okay, I guess we're getting married."

Jessica: I shouldn't laugh this hard, but I'm so laughing-

Carolynn: Oh, my. I thought you were going to say, will you marry me? I'd be so honored, but "Oh, I guess we'll get married."

Jessica: But then did you do a redo? Did you have a formal proposal?

Haya: There was no need to. There is a formal proposal by meeting my family and my dad and my mom and my brother, and they need to accept him so-

Amjad: That was a whole ordeal because I was unemployed.

Carolynn: Why?

Amjad: And broke.

Carolynn: You had an offer letter. You weren't really unemployed. You had an offer letter.

Amjad: Yeah, that's true. That's true.

Haya: He didn't have an income or a job. So that's [inaudible 00:14:30]-

Amjad: I had an offer letter from two kids in New York.

Jessica: Was the-

Haya: Correct-

Jessica: Offer letter mostly for equity or what was it that you were trying to increase, the cash or the equity-

Amjad: Yeah, the equity. I didn't care about the cash one bit. And that's a consistent theme in my life is that I just always was focused on building wealth. And a lot of what Paul talks about in his essays around getting rich and all of that, I just thought salary, cash, money is just like, yeah, whatever, that's what you need to live. And actually Zach did pay me very low. It was very painful. It was-

Carolynn: In New York especially-

Amjad: \$80,000. Yeah, \$80,000.

Carolynn: And this was like 2011. 2012 you said?

Amjad: Yeah, it wasn't-

Carolynn: Not that long ago-

Amjad: It was tough. I lived in a studio with a roommate. That's how bad it was.

Carolynn: Well, wait a second-

Jessica: Whoa, whoa, whoa, whoa, whoa.

Carolynn: Yeah, when they hired him-

Jessica: We have some questions here.

Carolynn: Yeah.

Jessica: Why were you with a roommate? I thought you guys... Wait, first of all, I need to find out just quickly-

Amjad: To get married and-

Jessica: Because I know we have a lot to get through. The parents, I need to find out what your parents thought, Haya, if they're shocked, he has no income, but you'd been together for a little while, right?

Haya: Yeah, we're talking to each other. So we-

Amjad: They didn't know that, that we were together.

Haya: No, no, no. They know that I talked to him as a colleague because we worked together in a startup. That's how we met. Yes.

Jessica: Okay. So they had known who he was. Okay.

Haya: Yes. And they knew that he was an engineer I worked with. My mom met him and liked his behavior. So I think my family and where we come from, we try to focus on values and ethics. Money comes and goes. There are things that don't come and go, which is passion, honesty, and how someone would treat their mom or their dad. And this is a projection of how they would treat their family, later on, their wife and their kids. And this is what we loved about Amjad. And he doesn't have a job today, he'll eventually have a job, he'll eventually have money and I am with him in the long run.

So this is what my dad saw and what my brother saw and my mom, and we just loved the family. Yeah.

Jessica: Okay. All right, so that worked out. So then, Amjad, you move to New York and you have a roommate?

Amjad: Yeah. So to get married, yes, all this talk about money and all that is all nice, but ultimately weddings in Jordan are expensive. It's an honor culture and you have to show face. And my father was in the government and he had a reputation to keep and all of that. So we had to pay for the wedding. And so I had to save quite a bit of money. And so I went to the States and I just wanted to save a big part of my salary so that I can go back and pay for a big wedding. And it was a big wedding. It was a huge wedding. And so to do that, I had a friend who was going to NYU, and so I called him and I said, "Hey, I'm coming to New York, where do you live? Should we room together?" And it was like, "Yeah, we could do that." So we started looking for apartments and I had an Excel sheet for how much money I needed to save and all of that.

And just, it didn't work any way to have a two-bedroom apartment or anything like that. So in the Upper West Side, actually the area was not too bad, but it was just one room. We had two beds next to each other, one bathroom, and we lived together there. But it was interesting, because he had this college life and he was going out with his friends and all that, and I was friends with them so I got to see that. But he would play World of Warcraft all night. And I'm trying to sleep. I have work at 9:00 AM and I hear mouse clicks, at night.

Jessica: This is not where I thought this interview was going to go. I'm a little thrown off. It's so fascinating.

Carolynn: Haya, you stayed back in Jordan. Did you have a web design job at this point? Were you consulting? What did you do during this time?

Haya: I was learning how to cook because-

Carolynn: Not what I thought you were going to say.

Haya: I did not have a job. I just got engaged. So I expect my significant other to take care of me now and I'm going to learn how to cook, which is a skill I'm not practicing now, but I still have my book with me. But yeah.

Carolynn: Okay.

Amjad: She took an intensive cooking course with her mom, who's a master cook, and she has this black book that she breaks out once every two, three years to cook something.

Haya: Yes.

Jessica: I need to do that. I don't cook very much. So then walk me through how much time passed before you got married and then, Amjad, you went to Facebook, what... Quickly bring us up to speed on that because I have to fill in these blanks.

Amjad: So everything happened super quick. It was actually dizzying speed. I remember telling Haya after everything was said and done, I was like, "Wow, I feel like I compressed 10 years of my life into one or two." So basically I went to the States, two months... Well, actually on my way to the States, I got an O-1 visa. Peter Roberts, YC's favorite lawyer, got me an O-1 visa. An O-1 visa is an alien of extraordinary ability just based on my open source work and all of that. On my way to the airport, actually, they didn't know what an O-1 visa was. I had saved \$700, \$800 and they made me pay for a ticket back because they didn't think it was an immigrant visa, they thought it was a visitor visa. And so when I landed in New York, I was so broke.

Carolynn: Wow.

Jessica: Oh my God-

Amjad: But you know-

Carolynn: Brutal.

Amjad: Yeah. And so yeah, I started building up some savings and all of that. Went back, did the engagement party in April and went straight back to work for another six months. And then in September that year, went back, got married, went to Thailand for a couple of weeks for the honeymoon, and then New York back to work. And then worked at Codecademy for another year and a half, I think at the time.

Late 2013, I just wanted to do something more technical to be honest. My job at Codecademy just became leading and hiring and doing all this kind of stuff that early employees do. And the company had grown. It was like 30, 40 people and I just felt I wanted a new challenge. So I went to... Facebook was doing really exciting work at the time. I was inspired by Zuck's vision around Internet.org, which was supposed to connect everyone in the world. He was talking a lot about the internet, it's this amazing thing, it's going to lift people out of poverty. A lot of people are going to get introduced to computers.

And my life story just showed me the power of the internet and the power of software and the power of things like open source and how democratized it's going to get, and the idea that people can really fundamentally change their lives if they learned a skill or two online. And so I was inspired by that mission. I joined Facebook and quickly got attracted back to developer tools because I tried to write a Java application for their Android platform and it was so slow. The whole process was slow. I was just like, "How come the tools are so bad," even at a big company like that.

And so there was a project at the time there that was trying to bring web development to mobile. It's called React Native. So it's trying to bring React JS to mobile. And I was

one of the founding engineers. I was on a team of three and now this is the top cross-platform application in the world. Discord's written in it, Airbnb, a bunch of apps. And it was really fun. It was a startup within a big company because we built this thing that reached millions of developers and billions of people because a lot of people were using these applications.

And it was until 2016 that I started getting the edge to do something new. And at the time, I'll let Haya take over from here because she revived the idea of Replit and the project that is Replit.

Jessica: Okay, Haya, how'd you revive this? Because I feel Paul had said, I talked to him briefly this morning about this interview, and one of the things he said was like, "Replit is the exact opposite of a pivot. This has been cooking and percolating for years. For years it was Amjad's life's work kind of thing." And so, Haya, how did you reinvigorate the idea?

Haya: So I joined Amjad in New York, so I thought that I would get a job-

Jessica: Because you're married now. Are you married now?

Haya: Yes, we're married-

Jessica: Okay, so you then joined him in New York. I just had to get closer on the timing. Okay.

Haya: Totally. Yeah, yeah. And then I thought just chasing the American dream, I'm in New York, I'm a designer. This is the best city to be a designer, and I thought if I apply to a designer job, I'll get a job immediately. But it was like pulling teeth. Very soon I realized I am competing with the best designers in the world, like designers from Stanford, Harvard, you name it, what makes me stand out? And I didn't know. And then I started looking at different portfolios and I felt super discouraged with my work.

So by talking to Amjad's friends and the designers in Codecademy and talking to Amjad, their recommendation was like, "Keep practicing." So I kept practicing and practicing and finding different gigs. And then I stumbled on this niche, which is internationalization design, finding gigs that have to do with the MENA region, like companies and products that try to focus on the Middle East and North Africa where they need cultural design or different design for their websites because the language is from right to left. So mirroring the design does not work. It needs to be an experience, like a different experience. It comes from ergonomically and how it functions or the feel of it or the direction or the narrative, even the color of the models or the pictures on the website. While that was like, okay, a dream coming true, working nice gigs on the biggest agencies in New York, I hated it. I was like, "Okay, the checkbox is done now. I just hate being a brand designer."

Amjad: I remember she used to tell me about how people would check out at 1 PM on Friday. Their weekend starts right after lunch. It's like, "Oh my God, these people don't work hard at all."

Jessica: Wow-

Amjad: They start drinking at lunch on Fridays and...

Haya: Yeah. And they would have this blame culture if the work does not happen, they just have a whole chain of email of why, whom, whom, whom. And then I told her and I'm like, "This is not working for me. I don't like it." So I started to think, "Maybe I need to go back to school and do a masters or what should I do?" And then back in the day it was up and coming, these boot camps, boot camp for UX, boot camp for engineering and general assembly, one of the hottest ones. And then Amjad told me, "I have my friend working at this boot camp. Maybe you just go and check it out, intro to." So I was like, "Sure. Yeah."

And we did not have the money to go back to school. So I went to the intro to UX/UI. And for the first time in my life, it's just all the things that I've been building towards, all the things that I've been talking to, all the things that I don't have language for, it just made sense. And then I came back home super pumped and told Amjad, "Hey, I want it, but we don't have the money."

So I asked my parents to lend me the money because I really wanted to be on this 10 week course. So I did the course, finished it, and then soon later on in tandem with working and Amjad on fun project, side projects that did not stop, I soon realized that in the 10-week course, we barely scratched the surface of every single topic and it's on me to deepen my knowledge with every single area.

In the boot camp, they prep you for a job, where most of my batch found a job, I couldn't find a job. For the first time I'm landing an interview, but I'm not getting an offer. So it was a super frustrating experience and very disarming. You come in with confidence and you start to feel less confident and questioning your talent.

So Amjad asked me to stop researching for a job for a whole year, and we did the research together. I remember very well, he was at the time, at Facebook, what does it take to be a designer in Facebook? Because Facebook at the time was hiring the best designers. So from the research we found that they're fully in, they're blog, they are photographers, they have side projects, they're all in design. It's their lifestyle. This is not something they do for a job. So he's like, "Listen, I really need you to stop looking for a job and work on a side project and it will help you all the way. Just think of any idea that you have."

And we started brainstorming, "Should we do a fitness app? Should we do a lifestyle app? Should we..." And then I'm like, "What happened to Replit? I see you sometimes maintaining the project on weekends. How about we work on it?" And he's like, "Are you serious? What do you want to do with Replit?" "It has users, yeah?" And he is like, "Yeah." "Well I want to talk to these users. I want to start from somewhere. This is what I learned from school: talk to users, do interviews, survey, understand." He was like, "Okay, do it." So started working with Amjad and bringing the users and doing surveys. And then I was so shocked by how much there was love for this one pager on the

internet that didn't even have any counts, that didn't even have... it's just one page. And I started thinking with Amjad, what do we do with it? What is the direction? And it could have gone so many ways. It could have gone nonprofit, it could have gone like a startup.

And I started living and breathing this thing for a whole year and I was so into it that I forgot that this is becoming a thing. And when I started going to interviews and I'm pointing at my work at Replit, it was an eye-opener in one interview where they said, "It looks like you have something working here. Do you really want this job?"

Jessica: Oh my gosh-

Carolynn: Wow-

Jessica: Someone said that to you in the interview?

Haya: Yes, and it was an epiphany moment that it was like an eye-opener. I left the offices just looking at the offices. I was like, "I don't want to work here." And I went back home and asked Amjad, "I think I really, really like working on Replit. So what do we do now?"

Jessica: Okay.

Haya: Yeah.

Jessica: And so were you ready mentally, Amjad, to make the leap?

Amjad: No, I saw what Zach and Ryan went through as founders. It was hard. It was really hard. And I had empathy because I was so early and I think it was hard on me as an early employee. I would multiply that by 10 to 100 for the founder versus the early employee in terms of stress and how much you have on the line and your reputation and all of that. Ben Harwood says it gets intertwined with your nervous system and the attacks on your stomach because almost it's a recipe for all sorts of disasters. So I thought I just didn't feel I wanted to do it. I was feeling good at the time. I got healthy, I was starting to work out and I was eating healthy because for a long time I was working really hard, neglecting my health and I was like, "Oh, I don't know if I want to do it."

So we tried to make it a project at Facebook, actually. I went to my manager, I was like, "Hey, I want to work on this, but I think it could have a great impact on the world and would make programming easier. It would unlock opportunities across the globe." And they were interested and we tried exploring that, but it just never panned out. They just felt I'm going to lose ownership of this thing and I'm going to have a boss and all of that. So-

Haya: But they offered me a job at Facebook.

Amjad: They did.

Haya: And it was-

Amjad: They did eventually. So we got-

Jessica: Really? Okay.

Amjad: The plan worked. Yeah. They were great. I think it's a fantastic company. And so I decided finally, after a lot of deep thinking. The thing that really Haya and I felt we care about is the mission. Until this day, making programming more accessible to more people is changing lives as we speak. A lot of people on Replit have fundamentally changed their lives because they learned the skill.

And so we took the plunge and I wrote a master plan. So the master plan, I was inspired by Elon at the time of this Tesla master plan. And the master plan, actually I have it up on the website somewhere, but basically the first step is we're going to make this tool. It's not going to be that powerful. So the best use case for it is to learn to program and we're going to sell it to schools. That's how we're going to monetize and teachers and students are going to use it.

Then the second step is we're going to improve it tremendously and make it so that you can do more things with it. We're going to have a lot of data. We're going to train an AI, and the AI is going to help you to code, and that way it's going to make programming even more accessible. And mind you, this is like 2015, '16, right, I'm writing this thing and AI was not what it is today? So I saw that it was going to come and it's going to have an impact on coding.

And the third step was like, oh, Replit is going to become a one-stop shop to build any source of software. You're going to be able to host the applications, you're going to be able to deploy them and scale them. You're going to be able to build an entire business on it. And so that's where we are today. So what Paul said, exactly is it was always that vision was always the case.

So when we applied to YC the first time, we're like, "Oh, we're kind of an education company. We have all these teachers and students we're going to sell to them." We got rejected, but we started bootstrapping essentially because we started getting some revenue. Eventually we raised some money from Bloomberg Beta from Roy Bahat who was also an investor at Codecademy. I knew him from those days, I reached out to him, but still it wasn't a lot. We raised half a million dollars, and so we were bootstrapping for a long time and we actually had a profitable month or two at some point until I got this DM from Sam Altman in late 2017.

So Sam sends me a DM and he's like, "Hey, I run YC." I'm like, "Dude, I know who you are. Come on." And he's like, "We've been looking at Replit and we think it's super interesting." I was like, "If it's super interesting, why didn't you accept us the gazillion times we applied?"

Carolynn: Good question.

Amjad: Yeah. And he's like, "Why don't you come see me at OpenAI?" I was like, "OpenAI, what is that?" And so I go to the mission and I see him at OpenAI and we started talking and he is like, "What you're building is super exciting. Actually, Paul found you in Hacker News," Hacker News at the time loved us and we were on Hacker News every other week, "And you should talk to him before you decide you want to do it or not". So PG and I started an email relationship. Because at the time, I think late 2017, he was very committed to the retirement idea, and I don't think he was getting on phone calls or any of that-

Jessica: And we had moved to England. Well, he doesn't get on phone calls ever-

Amjad: Anyways, right? I...

Jessica: And we were in England. Okay. But I knew he met you first and he has always loved this idea. In fact, he tried to build it, Carolyn, if you remember in the episode with PG, remember he tried to start his own company that was a combination of Replit, Heroku, and a programming language. And he failed-

Carolynn: Yeah, I do remember that.

Jessica: And just did the programming language, but this is always one of his pet ideas. So I think when he met you or talked to you, he was super excited and funded you, right?

Amjad: Not yet So he-

Jessica: Not yet? Okay.

Amjad: We talked for three months over email and I think I could print out a book of the essays he sent me. It was really great because for the first time someone got Replit, got it on a gut level, and we talked about all the things that are broken in the world of programming today and how we're going to be able to fix them. And then towards the end he was like, "There's a new YC batch starting. You should consider it." That was the only thing that he said. And so I went to Sam and I was like... Well, actually, I went to Haya first and I was like, "Hey, we have a business and do we want to give 7% to YC because we're already kind of off to the races." It wasn't actually that big.

And so the reason we decided to do YC is Paul, because we felt his advice was so on point. It felt like talking to a prophet in some ways because he could see the future so clearly. And I actually think Paul has a big reputation. Actually, after interacting with him personally and hearing his ideas, he's probably underrated, I would say, in terms of how clearly he can see things. And a lot of founders who would still meet with him today for Office Hours and all of that, he can drop one insight and that could really change the trajectory of your business.

Jessica: Yeah, I've seen that happen before.

Carolynn: And I think-

Amjad: Yeah, I'm sure, I'm sure-

Carolynn: He's going to forgive you for the Lisp comments if he hears all this. So I think you're good-

Jessica: Yes, definitely.

Amjad: And so Sam sends me... I tell Sam, "Hey, Paul asked us to do YC." And it's like, "Oh, that's a great idea." "Hey, this is just a formality. Do this application." I'm like, "Oh God, damn I don't want to do this application again." And-

Jessica: Is that why there was a freaking Rickroll in the video?

Amjad: Yes.

Jessica: Is that what you were leading up to? Okay. And that wasn't on my list of questions, but now it makes sense. When I was looking at this application, I was like, "Why is there a Rickroll in here?" I was so confused-

Haya: Yeah. Because we've done it-

Amjad: Yeah, that-

Haya: So many times, four times-

Jessica: How many times did you apply to YC? I thought just once before that-

Haya: No.

Amjad: No, there was another-

Haya: June-

Amjad: Couple, maybe different emails, something like that-

Jessica: Oh, no. Okay. Oh-

Haya: We applied at least three times. Yeah.

Jessica: Okay.

Amjad: Yeah.

Carolynn: Okay.

Amjad: So I wanted to get back at you all for it because [inaudible 00:41:03]-

Jessica: All right-

Carolynn: That is so funny.

Jessica: That's funny. Okay, so you went through the interview process though?

Amjad: Yeah, and Michael was livid. He hated the fact that we were crawled on the application and then we became friends and we're now best friends-

Haya: He was just... yeah-

Jessica: Oh, did he find that just too cheeky?

Amjad: He was like, "That's not how you get into YC," and you know Michael-

Haya: Offense, yeah.

Amjad: Yeah. When he's mad, he's mad, right? See I was nervous. I thought we're not going to get in. But yeah, we got in.

Jessica: Oh, I'm glad you did. Well, was YC useful having-

Amjad: Yes-

Jessica: Finally gotten in? Tell us any highlights.

Haya: I remember when we joined the speech that Sam gave to the batch at dinner, he said something that stuck in my head forever, "If you ever thought that you worked hard in your life, you are wrong. The next eight weeks will be the hardest times of your life. You're going to work so hard, you're going to love it, and then you're going to look back when you're done and you'll be so proud of yourself." I was like, "Bring it in, we're ready." And he was looking at us while he was also giving the speech and-

Jessica: Are you sure? Are you sure he was looking directly-

Haya: I'm sure. I remember, I'm sure. And Amjad was looking at me. He was like, "Why is he keep looking at us?" And he was like, "We're up to the..." Remember, Amjad? I'm not...

Amjad: Yeah.

Haya: I'm not... Yeah.

Jessica: When you were in YC though, you had a product, you had been making some money. So what was the big thing that you focused on during those intensive three months?

Amjad: Yeah, Jessica, the thing is we wanted to evolve the product. We never wanted it to just be an education product. Learning to code is great, and that's one of the reasons I left

Codecademy was that I just want to build a programming platform. The fact that it makes it easy to learn is great, but I just want people to build. I want people to make things. I want people to, again, change their lives. And YC gave us a license to go big, to finally go really big.

And during YC, the main project we had on our mind is, okay, now we want people to build web applications. We want people to be able to host applications. So that was the big thing. And-

Jessica: Ah, okay, okay.

Amjad: You have to understand, when we got into YC, Replit was basically one file editor and a console terminal output. And you couldn't even do games. You could do text-based games and things like that, which kids found really fun. But we didn't have multiple files. We didn't have a way to import libraries or projects. It wasn't just powerful.

I think in the eight weeks that we were working on it during YC, we evolved the product by leaps and bounds. By the time we were done with YC, we had hundreds of thousands of websites hosted on our applet because people were making, especially young hackers, like 14, fifteen-year-old kids were making a lot of websites. And for the first time in their lives, they were empowered to make these things.

And we also discovered that we had an amazing community. So when they were building these websites and apps, a lot of them were social. So Replit started having social features, chats and users were connecting with each other. They started working with each other. So we started running hackathons, and started doing meetups and things like that. So the product evolved in those eight weeks more than it would've taken us a couple of years.

Carolynn: Did you guys have help or was it just the employees meaning or was it just the two of you working on this?

Haya: The thing is with Amjad and I, the thing that we talk a lot about that we did not even start from zero. We start from minus. We don't even have connections. Most of the companies at the time when we were interviewing people to join us, most of the other companies, they would compare us to other companies that are doing less than us. So the prestige that we gained with YC was huge for recruiting and hiring and even just-

Jessica: Oh, good-

Haya: Having our name joining YC was really a big validation for recruiting and raising.

And when Sam gave that speech, we came back super pumped and he's like, "Hire everyone you know. It feels like we're going to war or something. Bring all your friends or your family, literally anyone who can help you."

Amjad: He said that? I don't remember he said that.

Haya: Yeah, yeah, he said that-

Amjad: Yeah. Okay.

Haya: Yeah. And then we came back to the office and we started thinking it was the three of us, Amjad, me, and our first engineer and we start-

Amjad: Who was an 18-year-old runaway from home kid.

Jessica: Oh, no-

Carolynn: Oh my God.

Haya: Yeah.

Amjad: Not really runaway but-

Haya: He's great-

Amjad: Yeah, he's awesome.

Haya: He's one of-

Amjad: He's a millionaire now and...

Haya: Yes. He's-

Jessica: Okay, good-

Haya: One of the smartest engineers. He is probably 25 now.

Amjad: Yeah.

Jessica: Haya, you've made me think of two things while you're describing this that are related. One, you were saying you had no connections. I feel the round you raised after YC was the most insidery Silicon Valley group of investors you could put together. I mean the total opposite of Theranos investors.

And the second thing was, did you try to recruit some of the best Replit users that you knew? Is that something you guys did?

Amjad: So that eighteen-year-old kid was a Replit user and that was-

Jessica: Ah, okay-

Amjad: Our first employee. And then when we got into YC, like Haya said, we really needed as many people as possible. So I flew my brother from Jordan who's still at the company, he never-

Jessica: Oh-

Amjad: Went back. He runs our IDE team, our editor team.

Carolynn: That's amazing-

Jessica: All right.

Haya: Yeah. And we literally started to think about every designer, every person, and I'm just thinking with Amjad, the people who worked with him-

Amjad: Yeah, I hired-

Haya: "Would you work with this person, would you work with this person?"

Amjad: So yeah, we put together a great team really quickly in-

Carolynn: Wow-

Amjad: A few days and we were on the warpath and we were really excited. I think the main thing, Jessica, that I think YC does to founders, in my opinion, is that just the energy it gives you, it's that just the entire aesthetic and the orange room and the way people talk and the way you just feel you're doing something important, that you're surrounded by a group of super smart people, you're kind of competing against them in terms of who's going to work hardest, who's going to get the most progress week over week. To us, that was really the most exciting thing about doing YC.

Jessica: Kicks things up a couple of notches.

Amjad: Yeah.

Jessica: One thing, again, I was talking with Paul about that he pointed out to me that I don't think I consciously thought about. He said, "Replit has large amounts of system software," like AWS or whatever, "But is aimed at consumers, which is unusual."

What's it like doing a startup that has that unique element to it and what was the hardest technological problem that you've had to overcome?

Amjad: Yeah, I mean we're building an entire programming stack from scratch. It's all these layers. It has an operating system layer and it has a programming interface layer, it has a package management layer, and all of this comes together in a very neat UI that hides all the complexity away. And so yeah, we have amazing talent today, amazing infrastructure talent.

And the cool thing about my relationship with Haya is that I'm very infrastructure-minded. I can understand what good design is, but she's the designer and the company always was this marriage of infrastructure and design-

Jessica: Yes, which Paul also mentioned was very unusual as well for a super system software type of company to have style. And that's you, Haya, brings the graphic design style. No offense, Amjad.

Amjad: No, that's totally true. Yeah.

Haya: I feel understanding, as in the beginning of the call, we're not technical and it's very important to think about everyone. People are non-technical, people are technical, people are kind of technical and we talk a lot about simple design. Simple design is the hardest thing possible to do. And something that also Paul talks about, if a five-year-old understands it and a seventy-year-old understands it, then you get everyone in the middle.

So sometimes when engineers have a common language or a common understanding of a concept, and they agree on all of it. But how can non-engineers also understand it? So it can be really simple and get the success of, "Hey, this is a lot easier than the other tools that you were using." Because the ease of use is what makes users want to use you versus other products in competition and...

Amjad: I just want to answer, you had a question about what's difficult about it technologically.

There are a lot of things around scalability, but I want to point out one which is security. Now many people think about Replit when they think about security, but when you're giving computers to random people on the internet, you're opening yourself up massively, right? It's literally-

Jessica: To fraud.

Amjad: To fraud, to hacking, to all sorts of malicious behavior.

And the thing I didn't know, I mean there's this famous Steve Jobs Stanford commencement speech where he talks about connecting dots backwards and the things you've worked on somehow influence your future career. And the thing I didn't expect is that I had a bit of a security background. I did a security set, I was a hacker kid. I used to love hacking into websites and things like that. And so I applied that knowledge and one of the things, we started getting competitors. And around the time we got into YC, we actually got a huge competitor in New York, it's called Glitch now that they got acquired. Or Joel Spolsky started a company that's like Replit and they got \$30 million in funding, there were 80 people. Because it was a spinoff from Fog Creek software, their existing software company.

And the thing that really made Replit stand out from a lot of the competition is that we were able to manage that security risk quite well. And so a lot of the competition were

either down all the time or were closing down their systems or not adding a lot of program languages and features because of the security issues.

Jessica: I love hearing that because one of the comments I have written down was that Y Combinator is like that in that it looks easy and makes people want to copy it, but they don't quite understand what makes it so special. It's just so interesting that the whole security stuff, people don't realize what a threat it is and you knew.

I also need to ask because it's very rare that we have a husband and wife team startup, cofounders. I want to know what it's like managing a startup with kids. How do you guys pull it off?

Amjad: I think in many ways doing a startup with your spouse is probably easier than doing a startup on your own while married because it's super hard for your spouse to understand how emotionally draining the endeavor is. And I think it ends up straining relationships quite a bit.

There are challenges to it, obviously. One challenge is, it's really hard, now we do better, but in the early days, let's say on the weekend I want to stop worrying about whatever we were worrying about, then Haya is worrying about it and she wants to talk about it. So it ping-pongs and we're always worried and it goes into a loop. So it is really hard to get any distance from the company.

I think VCs have a bias against or at least used to be I think, we've heard it a lot that they didn't like that we're a husband and wife. I don't know if you and Paul had that.

Jessica: I hate that so much because actually, I'm nodding my head this whole time because I totally agree with you. When Paul and I started Y Combinator and worked together for many years, it was so nice because we each understood what was going on in the other person's lives. And, Carolyn, you and John too, I'm sure.

Carolynn: Yep.

Jessica: And that's really nice. And actually a lot of the successful startups in history have been husband and wife teams. And so it's baloney when the VCs discriminate against married couples. Give me a break.

Amjad: Yes-

Carolynn: I think it's so weird too. I totally agree with you. I don't get it.

Amjad: Yeah.

Haya: I feel when we joined YC, part of joining YC is also that Paul and you, Jess, are married and it's like our role model. And for the first time I can reach out to someone who can understand me and ask you a question when I feel confused. Because we did have challenges with hiring people asking us questions like, "Do you make decisions at home?"

What if I want to be part of that decision?" So it was challenging to understand what is normal, what is not.

And I think it's also a second marriage to work in a startup together and be like cofounders, like are you in it? Are you going to leave if things go really, really rough? And I see this happening a lot, co-founders leaving each other because it's a competition. It starts to become a competition, who has the more-

Jessica: It can with a lot of people.

Haya: They just want more power or well, somethings gotta give here. If my other startup is the house and the kids and I'm going to be less involved in decision making and in the company, it should be okay because we have limited energy and we need to be okay with where our energy goes.

Amjad: I think the kids' part is interesting too. During the pandemic, we had our first kid and it was a crazy time for us. We were scaling like crazy. And we actually have pictures where times when we didn't have access to childcare, it was pretty rough in the pandemic, we would bring our kid to work. And actually Jared Friedman, who's our partner at YC, and I was talking to him yesterday, he asked me, "Can I share the story about the Replit house?" So our first office was a house and we spent a lot of time together. We cooked together, essentially lived together, played video games and poker and all that stuff together. And so when we had our kid, it was kind of okay, because we were living together and we had our kids as part of the company and all of that. So we integrated our lives and the company in a way that just made it work.

And I think over time as we got a little comfortable financially, that helps a ton, that really helps. Helps a lot. I think that the toughest period was when the company was growing really fast and we were still broke. That was tough.

Jessica: Yeah-

Carolynn: That's a lot of different pressure happening all at the same time.

Amjad: Right.

Jessica: All right, let's do one last question. Do people make money from Replit and who are they?

Amjad: Yeah, so we have a ton of people who, I mean especially the kind of people who want to be entrepreneurs who didn't have the skill of coding and either learned it with Replit or because AI made it easier, they dove deep.

One person, for example, Steve Marco, who last time I checked was on his way to becoming a millionaire, or I think he's in the hundreds of thousands of dollars all hosted on Replit. He runs this platform called Data.

Carolynn: Wow-

Amjad: And I think he's a photographer by background.

Jessica: Huh!

Amjad: And that's the coolest thing about Replit is that a lot of the people that are making money or starting businesses, we have another story of a woman, her name is Priya, she was a PM at Microsoft and she had a lot of ideas and she wanted Microsoft to build those ideas.

One of those ideas was can you generate a presentation from a document? Can you make a document and then just generate a presentation using AI? And she wanted to kill PowerPoint. That was her mission. And she still was doing that-

Carolynn: Okay. Worthy, that's a worthy cause-

Amjad: She couldn't get that done. Yeah. She couldn't get that done because she didn't get engineering resources at Microsoft. And so she started using Replit and she built it and then she quit her job, raised the seed round and she's off to the races.

Carolynn: Love that story. Love it-

Jessica: Oh, I love it. That's a great place to end then. I love that you're empowering so many people to change their lives. I love your mission. My fifteen-year-old has a Replit account, I think he probably-

Amjad: Amazing-

Jessica: Hopes that he can make some money off of it someday.

Amjad: Yes, he does, but yeah-

Carolynn: My almost-fifteen-year-old also has a Replit account, so-

Amjad: That's very cool.

Haya: That's awesome.

Carolynn: Yeah-

Haya: That's awesome-

Jessica: That's awesome. Thank you both for taking the time to come on today. It was really fun-

Carolynn: Yeah, thank you.

Amjad: Our pleasure. Happy to come back anytime and talk more about more recent things in the company because I know we spent a lot on the history.

Jessica: Yes, we did. I've got so fascinated by that. So thanks again. Hopefully we'll see you this summer.

Amjad: Yes. Yes.

Jessica: All right.

Amjad: We'll see you then-

Carolynn: Great to see both.

Jessica: Take care.

Carolynn: Bye.

Amjad: Bye.

Jessica: Bye.

Haya: Thank you-

Amjad: See you-

Haya: So much. Bye.

Amjad: Bye.

Jessica: Well that was really interesting, Carolynn.

Carolynn: It was fascinating and we spent so much time together because their personal life is really – their history together is so interesting. And I feel we could have gone on for so much longer-

Jessica: I know. I had no... we could have gone for so much longer. I have so many questions about their personal lives. We were like 40 minutes into it and we're like, "Okay, so you just started YC." I'm like, "Oh no, we're running out of time."

Carolynn: Yeah, well, you know what? We can bring people back for a part two and Replit's such a great company and our kids use it.

And also, you know what question I thought of? I was like, when he said he couldn't believe that this idea hadn't happened yet and I thought, "Well, is that gatekeeping? Do programmers not want other people to learn their craft?" And it's like, it doesn't sound

like it's that at all. It was more like no one had just really gotten around to doing it. Someone had to, it ended up being him. And everyone loves it. Everyone loves it.

Jessica: And by the way, it's really hard. It's one of those things, they make this easy user interface to make it easy for everyone. They're hiding all of the really hard stuff, which is incredible that they built all of this.

And I wanted to ask Haya about the community, because that's a really important element of Replit.

Carolynn: Yeah. And I also just think-

Jessica: Oh, well-

Carolynn: The fact that she brings this whole super deeply thought out design aspect to it just must make them leaps and bounds ahead of competitors. Because she's-

Jessica: Totally.

Carolynn: In it so deep, right, and she cares so much about that element of it. And she has so much expertise and I think that's pretty special.

Jessica: Yeah. And I loved learning more about it because I have to admit, this is the company that we've interviewed that I know the least about their background. Because of course, I was living in England when we funded them and they went through YC. So I was learning a lot. That might've been the problem, why we got so... we had to learn all the details leading up to the company because I didn't know a lot of it. It was fascinating.

Carolynn: Yeah, very, very fascinating.

Jessica: Oh, well I loved catching up with them and I'm so excited for how well Replit's doing. I wish them the best-

Carolynn: Yeah. That's awesome. That's awesome.

Jessica: And lastly, your hair looks really good today, C. Levy.

Carolynn: So it's because there's no humidity. I'm at altitude-

Jessica: Oh-

Carolynn: And it's zero humidity and so it's just flat and shiny. Thank you for noticing.

Jessica: Well, it looks glorious. So-

Carolynn: Well, thank you so much.

Jessica: Too bad the podcast listeners won't be able to see it.

Carolynn: Oh, I'm just assuming we're going to edit that right out of there.

Jessica: No. All right, Carolynn.

Carolynn: All right.

Jessica: I'll catch you on the flip side.

Carolynn: Have a great weekend.

Jessica: Bye.

Carolynn: Bye.